

Lab 1 Outline - Foundation

Descriptive Paper

1. Introduction

- 1.1 The beverage manufacturing industry generates billions of dollars annually and supports thousands of companies throughout the United States. Despite advances in manufacturing technology, many beverage producers continue to rely on fragmented processes involving spreadsheets, email chains, manual approvals, and institutional knowledge that is rarely documented. These inefficiencies create communication delays, increase operational costs, reduce visibility into production activities, and limit organizational scalability (Research and Markets, 2025). As beverage supply chains become increasingly complex, organizations require more effective methods for coordinating information and workflows across multiple stakeholders.
- 1.2 The main problem is the lack of a unified operational infrastructure capable of managing the numerous interactions between brands, suppliers, co-packers, logistics providers, and financial teams. An effective solution must provide centralized workflow coordination, real-time visibility into manufacturing activities, automated approval processes, document management, production tracking, and stakeholder communication. The system must also maintain a single source of truth, support data-driven decision-making, and provide scalable architecture capable of adapting to future business growth.
- 1.3 BevOS is a workflow coordination platform designed specifically for the beverage manufacturing industry. BevOS serves as an Operational Infrastructure as a Service (OlaaS) platform that centralizes communication, workflow management, production tracking, logistics coordination, and business documentation within a single system. Through the development of functional prototypes, this project will demonstrate how BevOS can improve operational efficiency, increase transparency, reduce manual effort, and provide a scalable foundation for modern beverage manufacturing operations.

2. BevOS Product Description

Provide a top-level description of CS 410 product for the average reader. Provide a summary of the solution – and its goals and objectives. This should be a one paragraph minimum.

BevOS is a system for connecting clients, suppliers, co-packers, freight, and finances as seamlessly as possible with a guiding principle that it is not the software that decides the system, the industry and its users do and it will follow them where they go.

2.1 Key Product Features and Capabilities

What does it do? What is significant/unique/innovative about it? What does it accomplish? Describe how it solves the problem.

2.2 Major Components (Hardware/Software)

Provide an overview of the hardware needed to support the solution. Describe how it is structured based on CS 410 MFCD. Define and describe the software to be developed.

3. Identification of Case Study

For whom is this product being developed? Why? Who else might use this in the future?

4. Glossary

BevOS: Workflow coordination software platform being developed for Nice Package Co.

Brand: The intangible sum of a company's reputation, values, and identity.

Client: Organization or individuals purchasing the service or products of another member of the overall manufacturing industry as part of their own business operations pipeline.

Co-Packers: Contract packager, a company that packages products for clients as a service to cover the enclosing and protecting products for distribution, storage, sale, and use.

Nice Package Company: Industrial coordination company working between internal customers to assist them in coordinating the beverage manufacturing and distribution process. Client for development of BevOS.

5. References

Research and Markets. (2025). Beverage manufacturing market research report. Retrieved from

<https://www.researchandmarkets.com/report/united-states-beverage-manufacturing-market>

Willson. Travis. (2026, May 21). BevOS [PowerPoint slides]. Canvas@ODU.

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Zimmerman, S. (2026, May 26). How the food and beverage industry is rethinking manufacturing investments. Food Dive.
<https://www.fooddive.com/news/food-beverage-manufacturing-investments-us-expansion/821081/>